

Title: Seasonal Variation Mediates the Importance of Species Attributes in Plant-Pollinator Interactions

Authors and mails: Jose B. Lanuza^{1,2} (barragansljose@gmail.com), Alfonso Allen-Perkins³ (alfonsoallen-perkins@gmail.com), Will Glenny¹ (willglenny@gmail.com), Anna Traveset² (atraveset@csic.es), Henriette F. Morgenroth¹ (hfmorgenroth@gmail.com), Raymond Umazekabiri¹ (Raymond.Umazekabiri@monash.edu), Marc Hoffmann¹ (hoffmann.marc@fn.de), Panagiotis Theodorou^{1,4} (panatheod@gmail.com), Robert J. Paxton^{1,4} (robert.paxton@zoologie.uni-halle.de), Isabell Hensen^{1,5} (isabell.hensen@botanik.uni-halle.de), Robert Rauschkolb^{1,6,7} (phenobs@uni-jena.de), Christine Römermann^{1,6,7} (christine.roemermann@uni-jena.de), Oliver Schweiger^{1,8} (oliver.schweiger@ufz.de) and Tiffany Knight¹ (tiffany.knight@idiv.de)

Affiliations: ¹ German Centre for Integrative Biodiversity Research (iDiv) Halle–Jena–Leipzig, Puschstr. 4, 04103 Leipzig, Germany. ² Mediterranean Institute for Advanced Studies (IMEDEA, UIB-CSIC), Mallorca, Balearic Islands, Spain. ³ Departamento de Ingeniería Eléctrica, Electrónica, Automática y Física Aplicada, ETSIDI, Universidad Politécnica de Madrid, Madrid, Spain. ⁴ Institute for Biology; General Zoology, Martin Luther University Halle-Wittenberg, Halle (Saale), Germany. ⁵ Institute of Biology/Geobotany and Botanical Garden, Martin Luther University Halle-Wittenberg, Halle (Saale), Germany. ⁶ Institute of Biodiversity, Ecology and Evolution, Friedrich Schiller University Jena, Germany. ⁷ Senckenberg Institute for Plant Form and Function (SIP), Jena, Germany. ⁸ UFZ – Helmholtz Centre for Environmental Research, Department of Community Ecology, Halle, Germany

Running title: Seasonal drivers of plant-pollinator interactions

Keywords: Plant-pollinator interactions; Species attributes; Species abundance; Trait matching; Phenology; Temporal resolution; Seasonal variation

Word count: Abstract, 190 words; main text, 4989 words.

Number of references: 76

Number of figures and tables: 4 figures and 0 tables.

Corresponding author: Jose B. Lanuza, Mediterranean Institute for Advanced Studies (IMEDEA, UIB-CSIC), Mallorca, Balearic Islands, Spain; barragansljose@gmail.com; +34 616 375 981

Statement of authorship: J.B.L., P.T., R.J.P., I.H., R.R., C.R., O.S., and T.K. conceived the study. J.B.L., H.F.M., and R.U. collected the data. M.H. identified the insect specimens. J.B.L. and A.A.-P. performed the analyses. J.B.L. wrote the first draft of the manuscript. All authors contributed substantially to the interpretation of the results, critically revised the manuscript, and approved the final version for publication.

Data accessibility statement: All code and data supporting the findings of this study are openly available at <https://doi.org/10.5281/zenodo.21559439>.